

Assignment 18 (ELEC 341 L18_NyquistCriterion)

Problem 1:

Draw Nyquist plot for the given $G(s)$:

$$G(s) = \frac{20(s^2 + s + 0.5)}{s(s+1)(s+10)}$$

$$s^3 + 11s^2 + 10s$$

$$G(j\omega) = \frac{20[(j\omega)^2 + j\omega + 0.5]}{(j\omega)^3 + 11(j\omega)^2 + 10j\omega}$$

$$j^3 = -j$$

$$= \frac{20(-\omega^2 + j\omega + 0.5)}{-j\omega^3 - 11\omega^2 + 10j\omega} \rightarrow \frac{-20\omega^2 + 20j\omega + 100}{10 - 20\omega^2 + 20j\omega}$$

$$= \frac{[(10 - 20\omega^2) + 20j\omega]}{-11\omega^2 + j(10\omega - \omega^3)} \cdot \frac{[-11\omega^2 - j(10\omega - \omega^3)]}{[-11\omega^2 - j(10\omega - \omega^3)]}$$

$$(a^2 - b^2) = (a+b)(a-b)$$

$$a = -11\omega^2$$

$$b = 10\omega - \omega^3$$

$$L(j\omega) = \frac{-20j\omega^5 + 200\omega^4 - 10j\omega^3 + 90\omega^2 - 100j\omega}{\omega^6 + 101\omega^4 + 100\omega^2}$$

$$h \text{ Re}(L(j\omega)) = \frac{200\omega^2 + 90}{\omega^4 + 101\omega^2 + 100}$$

$$f \text{ Im}(L(j\omega)) = -\frac{20\omega^4 + 10\omega^2 + 100}{\omega^5 + 101\omega^3 + 100\omega}$$

ω	0	∞	2	10	5	0.5	1.5	1.0
$\text{Re}(L)$	0.9	0	1.7115	≈ 1	1.5667	1.12	1.62	1.436
$\text{Im}(L)$	∞	0	0.4423	≈ -1	-0.99	-1.66	-0.449	-0.6435

Set $h = 0$, no Re intercept

$$200\omega^2 + 90 = 0, \text{ not pos.}$$

Set $f = 0$.

$$20\omega^4 + 10\omega^2 + 100 = 0$$

$$\omega^4 + 0.5\omega^2 + 5 = 0, x = \omega^2$$

no Im intercept.

